

Carlos Moreno Rojas

Biomedical Engineering · Electrical Engineering · Research & Development

PROFILE

Engineer with interdisciplinary training in Biomedical Engineering and Electrical Engineering, with experience in biomedical research, instrumentation, electronics, embedded systems, biomedical optics, medical imaging and experimental validation.

EDUCATION

MSc Biomedical Engineering — Pontificia Universidad Católica de Chile — 2025–2027

Research focused on ex vivo liver perfusion, biomedical instrumentation and ICG-NIR functional assessment.

Electrical Engineering — Pontificia Universidad Católica de Chile — 2025–2026

Electronics, embedded systems, data acquisition, control, FPGA and hardware/software integration.

Biomedical Engineering — Pontificia Universidad Católica de Chile — 2021–2024

Bachelor's Degree in Engineering Sciences — Biomedical Engineering.

RESEARCH & DEVELOPMENT

Master's Research — Liver Viability & ICG-NIR

Ex vivo liver perfusion platform; pressure, flow and temperature monitoring; vascular resistance; calibration and validation; physiological baseline; ICG-NIR fluorescence kinetics; functional parameter extraction.

Borealis UC — Organ Supercooling

Organ preservation research focused on supercooling and thermal control.

Organ Preservation Machine

Electromechanical redesign, fluid systems, control and miniaturization.

SELECTED ENGINEERING EXPERIENCE

CYBATHLON 2024 — Ciervo UC

Electrical Area Leader. Development and integration of electronics for a robotic transfemoral prosthesis.

Candel Medical Company

Professional internship in hardware, electronics and firmware for transcranial electrical stimulation.

Lung Segmentation — U-Net

Medical image segmentation using deep learning.

TECHNICAL SKILLS

Biomedical instrumentation; organ perfusion and preservation; NIR fluorescence; electronics; ESP32; STM32; Arduino; FPGA/VHDL; Python; C++; signal processing; medical imaging; CAD; Fusion 360; 3D printing; experimental validation.

TEACHING & LEADERSHIP

Physics Laboratory Teaching Assistant

Faculty of Physics UC — FIS154.

Biomedical Engineering Student Chapter UC

President, 2024–2025.

IEEE EMBS PUC Chile

Webmaster since 2024.

LANGUAGES

Spanish

Native.

English

B2 / ALTE 3.